

Exploratory Data Analysis (EDA) Project Report on the Iris Dataset

Introduction

Exploratory Data Analysis (EDA) is the process of analyzing datasets to understand their characteristics, identify patterns, detect relationships, and summarize important information through statistical methods and visualizations.

In this project, the Iris dataset was analyzed using Python and Google Colab.

Objectives

- Analyze the Iris dataset.
- Explore the structure of the dataset.
- Generate statistical summaries.
- Verify missing values.
- Create visualizations.
- Identify relationships between different features.

Tools and Technologies

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib

Dataset Description

The Iris dataset contains measurements of three species of iris flowers.

Dataset Features

Feature	Description
SepalLength	Length of the sepal
SepalWidth	Width of the sepal
PetalLength	Length of the petal
PetalWidth	Width of the petal
Species	Iris flower species

Dataset Size

- Rows: 150
- Columns: 5

Data Exploration

The following analyses were performed:

- Dataset preview
- Dataset shape
- Dataset information
- Statistical summary
- Missing value analysis

Missing Values

The dataset contained no missing values.

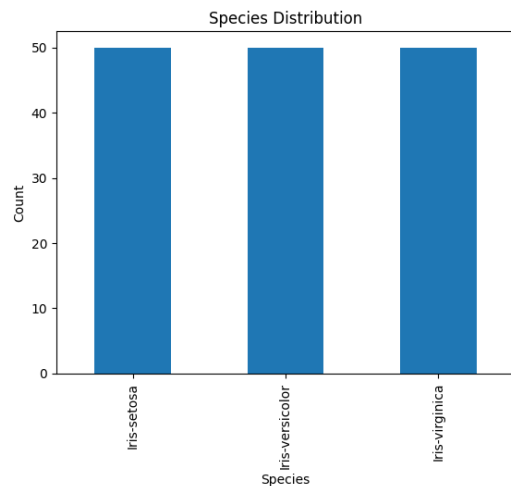
Data Cleaning

No data cleaning was required because the dataset was already clean.

Visualizations

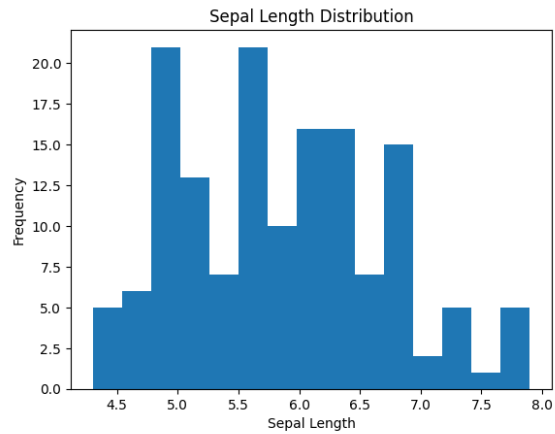
Species Distribution:

A bar graph was created to compare the number of flowers in each species.



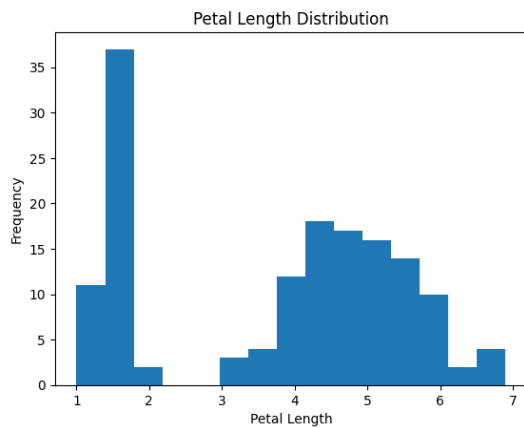
Sepal Length Distribution:

A histogram was created to analyze the distribution of sepal lengths.



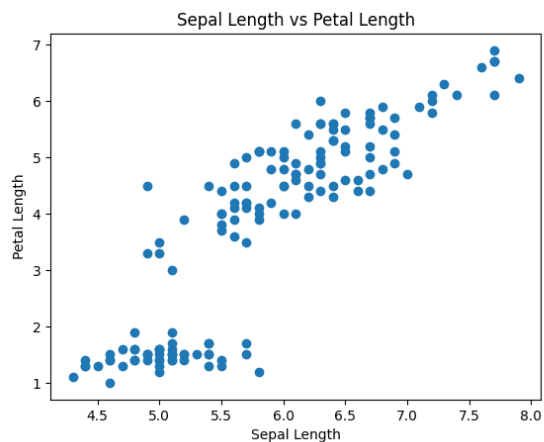
Petal Length Distribution:

A histogram was created to analyze the distribution of petal lengths.



Sepal Length vs. Petal Length:

A scatter plot was created to study the relationship between sepal length and petal length.



Findings

- The dataset contains 150 flower samples.
- No missing values were found.
- The three species are equally represented.
- Petal measurements differ significantly among species.
- Sepal and petal measurements show noticeable relationships.

Conclusion

Exploratory Data Analysis helped identify patterns and relationships in the Iris dataset. The project improved data analysis, visualization, and interpretation skills using Python.